Math Bootcamp

Day 4 — Optimization

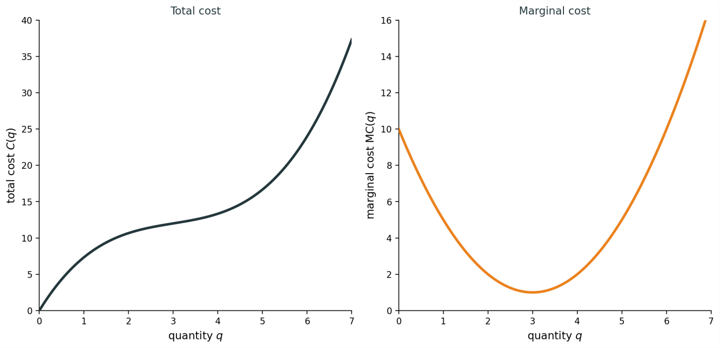
Math Camp 2026

Day 4 roadmap

1. **4.1** Unconstrained optimization — FOC and SOC
2. **4.2** Concavity, convexity, quasiconcavity — global optimality
3. **4.3** Equality constraints — Lagrange multipliers
4. **4.4** Envelope theorem — value functions
5. **4.5** Inequality constraints — Farkas lemma and KKT
6. **4.6** Why equality multipliers are unrestricted

4.1 A firm choosing output

A competitive firm chooses $q \geq 0$ at $p = 5$, with $C(q) = \frac{1}{3}q^3 - 3q^2 + 10q$.



Profit is $\pi(q) = 5q - C(q)$. What quantity maximizes profit?

4.1 The optimization problem

Consider: $\max_{x \in (a,b)} f(x)$ where f is differentiable.

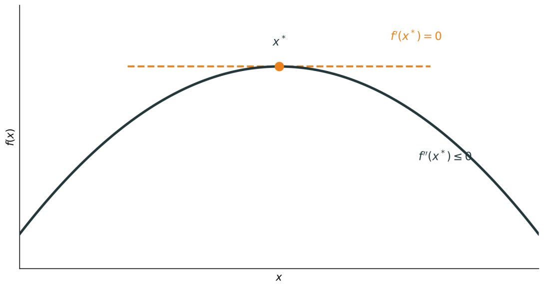
Goal: Find conditions guaranteeing x^* is a local maximum.

Two types of conditions (always say **for what**):

- **Necessary for local max:** Must hold at any interior local max, but does not by itself guarantee a local max
- **Sufficient for local max:** If they hold, x^* is definitely a local max (but need not be the only way to get one)

4.1 Necessary conditions at a local max

If x^* is an interior local maximum, the graph looks like this:



The tangent is flat, so $f'(x^*) = 0$. The curve bends down, so $f''(x^*) \leq 0$.

4.1 Local conditions

At an interior critical point x^* ($f'(x^*) = 0$) with $f \in C^2$:

Necessary (1). Local maximum $\Rightarrow f'(x^*) = 0$. **Not sufficient.**

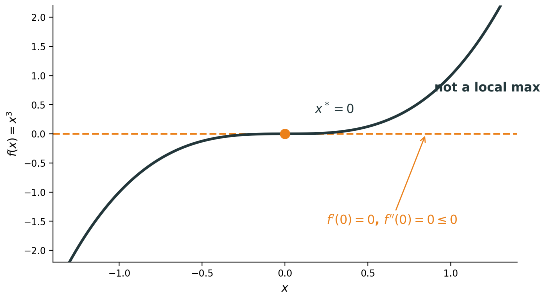
Necessary (2). Local maximum $\Rightarrow f''(x^*) \leq 0$. **Not sufficient** (e.g. x^3 at 0).

Sufficient. If $f'(x^*) = 0$ and $f''(x^*) < 0$, then strict local maximum. **Not necessary** (e.g. $-x^4$ at 0).

Proofs: Appendix A.1 **for interest only**

4.1 $f'' \le 0$ is not sufficient

$f(x) = x^3$ at $x^* = 0$: $f'(0) = 0$ and $f''(0) = 0 \le 0$, but this is not a local maximum.



Exercise. For the firm, $\pi(q) = 5q - C(q) = -\frac{1}{3}q^3 + 3q^2 - 5q$ on $q \geq 0$. Find all critical points and classify each.

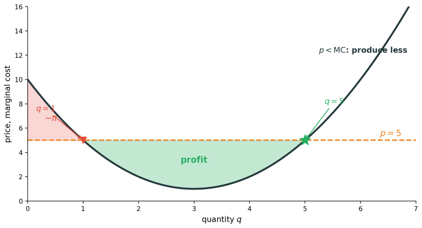
Solution.

- $\pi'(q) = 5 - \mathrm{MC}(q) = -q^2 + 6q - 5 = -(q-1)(q-5) = 0$ at $q = 1, 5$
- $\pi''(q) = -2q + 6$, so $\pi''(1) = 4 > 0$ — **local minimum**
- $\pi''(5) = -4 < 0$ — **local maximum** (sufficient SOC)

FOC gives two $p = \mathrm{MC}$ points; SOC keeps only the peak at $q = 5$.

4.1 Exercise: Why $p = \mathrm{MC}$ Together

Exercise. Why must $p = \mathrm{MC}$ at an interior profit maximum?

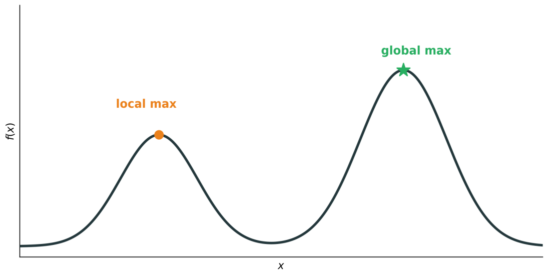


Solution. One more unit earns p and costs MC. Profit is the signed area between p and MC: add area where $p > \mathrm{MC}$, subtract where $p < \mathrm{MC}$.

Expand where $p > \mathrm{MC}$, cut where $p < \mathrm{MC}$, so $p = \mathrm{MC}$. SOC keeps the rising-MC point $q = 5$.

4.1 From local to global

A local maximum need not be a global maximum:



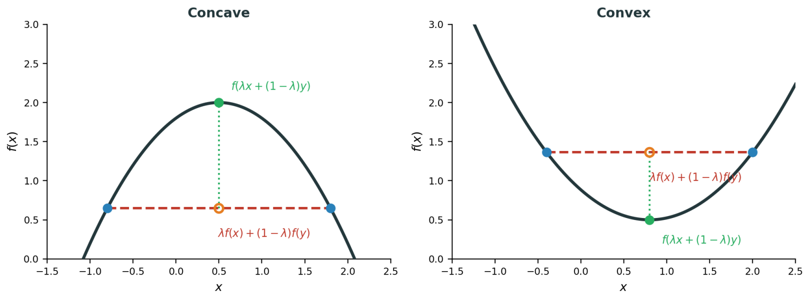
Question: When does a critical point guarantee a **global** maximum?

Answer: When f is **concave**.

(1) Concave. For all x, y and $\lambda \in [0,1]$:

$$f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda)f(y).$$

The graph lies **above** its chords (convex reverses the inequality).

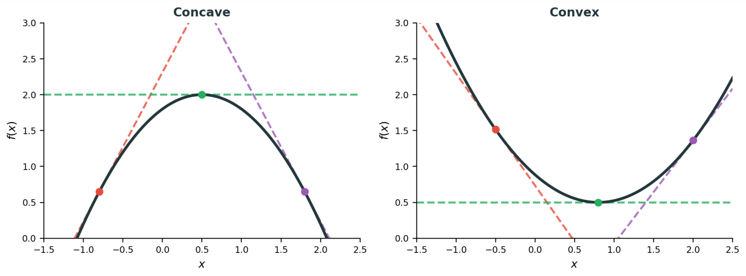


4.2 (2) Tangent characterization

(2) Concave (differentiable). For all x, y:

$$f(y) \leq f(x) + f'(x)(y - x).$$

Theorem. For $f \in C^1$: (1) $\Leftrightarrow$ (2).

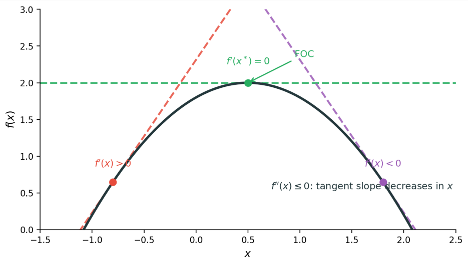


Proof: Appendix A.2 **for interest only**

4.2 (3) Second derivative condition

(3) Concave (C^2). $f''(x) \leq 0$ for all x — tangent slope $f'(x)$ is **decreasing**.

Theorem. For $f \in C^2$: (2) $\Leftrightarrow$ (3).



At a maximum, FOC gives $f'(x^*) = 0$: the tangent is **horizontal**.

Proof: Appendix A.3 **for interest only**

4.2 Exercise: Concavity implies global max Together

Exercise. Prove that if f is concave and $f'(x^*) = 0$, then x^* is a global maximum.

Solution.

- By tangent characterization: $f(y) \leq f(x^*) + f'(x^*)(y - x^*)$ for all y
- Since $f'(x^*) = 0$: $f(y) \leq f(x^*)$ for all y
- Therefore x^* is a global maximum

Concavity is **sufficient for global max** once FOC holds (FOC is only necessary for local max).

4.2 Concavity: Global condition

Optimization. When f is concave, every local maximum is a global maximum. In particular, if x^* satisfies $f'(x^*) = 0$, then x^* is a global maximum: $f(y) \leq f(x^*)$ for all y.

Three equivalent forms ($f \in C^2$):

1. **Chords:** $f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda)f(y)$
2. **Tangents:** $f(y) \leq f(x) + f'(x)(y-x)$
3. **Second derivative:** $f''(x) \leq 0$

Geometry. Graph above chords, below tangents; Taylor **overestimates** f.

4.2 Convex functions: Global minimum

Optimization. When f is convex, every local minimum is a global minimum. In particular, if x^* satisfies $f'(x^*) = 0$, then x^* is a global minimum: $f(y) \geq f(x^*)$ for all y.

Three equivalent forms ($f \in C^2$):

1. **Chords:** $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$
2. **Tangents:** $f(y) \geq f(x) + f'(x)(y-x)$
3. **Second derivative:** $f''(x) \geq 0$

Geometry. Graph below chords, above tangents; Taylor **underestimates** f.

4.2 Strict concavity

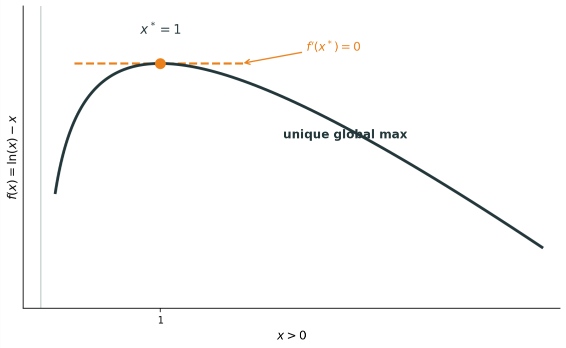
Optimization. When f is strictly concave, there is at most one global maximum. If $f'(x^*) = 0$, then x^* is the unique global maximum.

Three characterizations ($f \in C^2$):

1. **Strict chords:** for $x \neq y$, $\lambda \in (0,1)$: $f(\lambda x + (1-\lambda)y) > \lambda f(x) + (1-\lambda)f(y)$
2. **Strict tangents:** for $x \neq y$: $f(y) < f(x) + f'(x)(y-x)$
3. **Second derivative:** $f''(x) < 0$ for all x — **sufficient but not necessary**

Example. $f(x) = -x^4$: strictly concave, but $f''(0) = 0$.

Exercise. For $f(x) = \ln(x) - x$ on $x > 0$, find critical points, verify SOC, and determine whether x^* is a global maximum.



Solution.

- $f'(x) = \frac{1}{x} - 1 = 0$ gives $x^* = 1$
- $f''(1) = -1 < 0 \Rightarrow$ strict local maximum
- $f''(x) = -\frac{1}{x^2} < 0$ for all $x > 0$, so f is strictly concave
- Therefore $x^* = 1$ is the unique **global** maximum

4.2 Exercise: Testing concavity vs. convexity Together

Exercise. Determine if these functions are concave, convex, or neither:

(a) $f(x) = -x^2$ on $\mathbb{R}$

- $f''(x) = -2 < 0$ for all $x \Rightarrow$ **strictly concave**

(b) $f(x) = e^x$ on $\mathbb{R}$

- $f''(x) = e^x > 0$ for all $x \Rightarrow$ **strictly convex**

(c) $f(x) = x^3$ on $\mathbb{R}$

- $f''(x) = 6x$ changes sign at $x = 0 \Rightarrow$ **neither** concave nor convex on all of $\mathbb{R}$

4.2 Example: Profit maximization

Example. Profit $\pi(q) = pq - C(q)$.

- **Decreasing returns:** marginal cost rises as output increases (e.g. $C(q) = q^2$)
- Then $\pi''(q) = -C''(q) < 0$, so π is concave
- FOC $\pi'(q^*) = 0$ gives the **global** profit maximum, not just a local one

4.2 Local condition (multivariate)

At an interior critical point $\mathbf{x}^*$ ($\nabla f(\mathbf{x}^*) = \mathbf{0}$) with $f \in C^2$:

Necessary (1). Local maximum $\Rightarrow \nabla f(\mathbf{x}^*) = \mathbf{0}$. **Not sufficient.**

Necessary (2). Local maximum $\Rightarrow H_f(\mathbf{x}^*)$ negative semidefinite. **Not sufficient** (e.g. x^3 at $(0, 0)$).

Sufficient. If $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $H_f(\mathbf{x}^*)$ negative definite, then strict local maximum. **Not necessary** (e.g. $-(x^4 + y^4)$ at $(0, 0)$).

Same pattern as 1D: $f''(x^*) \leq 0$ necessary; $f''(x^*) < 0$ sufficient but not necessary.

Proofs: Appendix A.5 **for interest only**

4.2 Global condition (multivariate)

Optimization. When f is concave, every local maximum is a global maximum. In particular, if $\nabla f(\mathbf{x}^*) = \mathbf{0}$, then $\mathbf{x}^*$ is a global maximum: $f(\mathbf{y}) \leq f(\mathbf{x}^*)$ for all $\mathbf{y}$.

Three equivalent forms ($f \in C^2$):

1. **Chords:** $f(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}) \geq \lambda f(\mathbf{x}) + (1-\lambda)f(\mathbf{y})$
2. **Tangents:** $f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top(\mathbf{y} - \mathbf{x})$
3. **Hessian:** $H_f(\mathbf{x})$ is NSD for all $\mathbf{x}$ (recall, Day 3)

Geometry. For $f(x, y)$, the surface $z = f(x, y)$ lies **below** its tangent plane; shape is an inverted bowl.

(1) $\Leftrightarrow$ (2): A.2 **for interest only** (2) $\Leftrightarrow$ (3): A.4 **for interest only**

4.2 Negative semidefinite

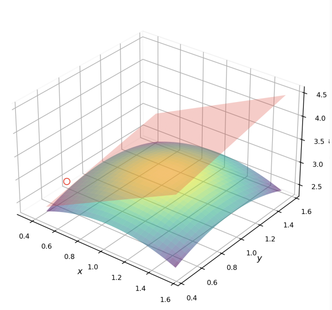
Recall (end of Day 3). For symmetric $H \in \mathbb{R}^{n\times n}$, the following are equivalent (proved on Day 3):

- **Quadratic form:** $\mathbf{v}^\top H\mathbf{v} \leq 0$ for all $\mathbf{v} \in \mathbb{R}^n$
- **Eigenvalues:** every $\lambda_i \leq 0$
- **Diagonal entries:** $H_{ii} \leq 0$ for all i (necessary); if H is diagonal, NSD $\Leftrightarrow$ all $H_{ii} \leq 0$
- **Leading principal minors:** $(-1)^k\Delta_k \geq 0$ for $k = 1, \dots, n$

$n = 2$. $H = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is NSD iff $a \leq 0$, $c \leq 0$, and $ac - b^2 \geq 0$.

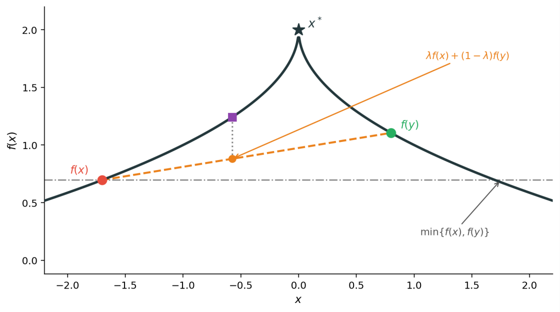
4.2 Multivariate concavity: Geometry

For $f(x, y)$, the graph $z = f(x, y)$ is a **surface in 3D**.



Concavity: surface lies **below** its tangent plane; shape is an inverted bowl.

Concavity gives a unique peak, but it is **too strong**: a single-peaked f can fail the chord test.



The graph can lie **below** the chord (not concave) and still stay above $\min\{f(x), f(y)\}$ — one peak, still good for maximization.

4.2 Quasiconcave functions: Two definitions

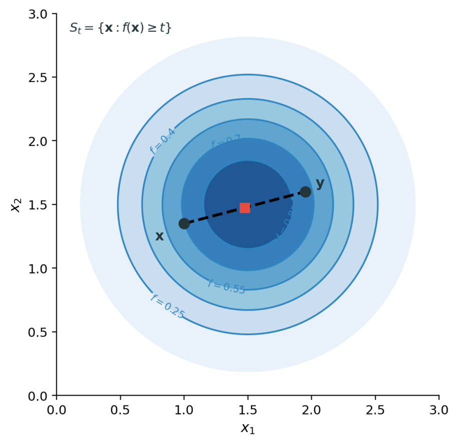
Definition A (segment). f is **quasiconcave** if for all x, y and $\lambda \in [0,1]$:

$$f(\lambda x + (1-\lambda)y) \geq \min\{f(x), f(y)\}.$$

Along any segment, f stays at least as high as the lower endpoint.

Definition B (upper contours). f is quasiconcave if upper contour sets $S_t = \{\mathbf{x} : f(\mathbf{x}) \geq t\}$ are convex for all t.

4.2 Two-dimensional upper contour sets



4.2 Equivalence: segment vs. upper contours

Theorem. Definitions A and B are equivalent.

A: $f(\lambda x + (1-\lambda)y) \geq \min\{f(x), f(y)\}$.

B: $S_t = \{\mathbf{x} : f(\mathbf{x}) \geq t\}$ is convex for every t.

Proof (A $\Rightarrow$ B). Let $\mathbf{x}, \mathbf{y} \in S_t$.

- Then $f(\mathbf{x}) \geq t$, $f(\mathbf{y}) \geq t$
- For $\lambda \in [0,1]$: $f(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\} \geq t$
- The segment stays in S_t, so S_t is convex

Proof (B $\Rightarrow$ A). WLOG $f(\mathbf{x}) \geq f(\mathbf{y})$.

- Then $\mathbf{x}, \mathbf{y} \in S_{f(\mathbf{y})}$
- By convexity of $S_{f(\mathbf{y})}$: the segment is in $S_{f(\mathbf{y})}$
- So $f(\text{segment}) \geq f(\mathbf{y}) = \min\{f(\mathbf{x}), f(\mathbf{y})\}$

4.2 Quasiconcavity and monotone transforms

Theorem. If f is quasiconcave and $\varphi : \mathbb{R} \to \mathbb{R}$ is **strictly increasing**, then $g = \varphi \circ f$ is quasiconcave.

Proof (upper contours). For any t, since φ is increasing, φ^{-1} exists and

$$\{\mathbf{x} : g(\mathbf{x}) \geq t\} = \{\mathbf{x} : f(\mathbf{x}) \geq \varphi^{-1}(t)\},$$

which is convex because f is quasiconcave.

Example. Quasiconcave utility u stays quasiconcave under $\ln u$, $u^{1/2}$, etc. — same indifference curves, same optimization.

4.2 Concave implies quasiconcave

Theorem. Concave $\Rightarrow$ quasiconcave, but not conversely.

Proof. For concave f and any x, y, $\lambda \in [0, 1]$:

- By concavity: $f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda)f(y)$
- WLOG $f(x) \geq f(y)$, so $\min\{f(x), f(y)\} = f(y)$
- Then: $\lambda f(x) + (1-\lambda)f(y) \geq f(y) = \min\{f(x), f(y)\}$
- Therefore: $f(\lambda x + (1-\lambda)y) \geq \min\{f(x), f(y)\}$

This is the definition of quasiconcavity. The converse fails: e.g., sigmoid is quasiconcave but not concave.

4.2 Quasiconvex functions

Definition. f is **quasiconvex** if for all x, y and $\lambda \in [0, 1]$:

$$f(\lambda x + (1-\lambda)y) \leq \max\{f(x), f(y)\}.$$

Equivalently: lower contour sets $\{\mathbf{x} : f(\mathbf{x}) \leq t\}$ are convex.

Key relationship: f quasiconcave $\Leftrightarrow$ $-f$ quasiconvex.

Example: $f(x) = x^2$ is quasiconvex (lower sets $[-a, a]$ are convex).

4.2 Log-concave and hierarchy

Definition. f is **log-concave** if $\ln f$ is concave (for $f > 0$).

Hierarchy (strongest to weakest):

1. Concave
2. Log-concave
3. Quasiconcave

Relationships:

- Concave and positive $\Rightarrow$ log-concave
- Log-concave $\Rightarrow$ quasiconcave

See Appendix A.6 for proofs **for interest only**

4.2 Economic examples

Cobb–Douglas: $u(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha}$

- **Not concave** for $\alpha \in (0, 1)$: the Hessian is not NSD everywhere
- **Log-concave** and **quasiconcave** for all $\alpha \in (0, 1)$

CES: $u = (x_1^{\rho} + x_2^{\rho})^{1/\rho}$

- Log-concave for certain parameter ranges
- Quasiconcave more generally

Why this matters: FOC gives global max on convex budget set.

Exercise. Show $u(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha}$ is quasiconcave on $\mathbb{R}^2_{++}$.

Solution. Definition B: show $S_t = \{\mathbf{x} \in \mathbb{R}^2_{++} : u(\mathbf{x}) \geq t\}$ is convex for every t.

- If $t \leq 0$: $S_t = \mathbb{R}^2_{++}$, convex
- If $t > 0$: $S_t = \{\mathbf{x} : g(\mathbf{x}) \geq \ln t\}$ where $g = \ln u = \alpha \ln x_1 + (1-\alpha) \ln x_2$
- g is concave: $H_g = \mathrm{diag}(-\alpha/x_1^2, -(1-\alpha)/x_2^2)$ is negative definite
- $\{\mathbf{x} : g(\mathbf{x}) \geq \ln t\}$ is an **upper level set** of g; recall: upper level sets of a concave function are convex
- So S_t is convex

Therefore u is quasiconcave.

4.2 Exercise: Cobb–Douglas is not concave Together

Exercise. Show $u(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha}$ is **not** concave for $\alpha \in (0, 1)$.

Solution. Concavity requires $H_u(\mathbf{x})$ NSD at every $\mathbf{x} \in \mathbb{R}^2_{++}$.

- Hessian entries (general α):

$$H_{11} = \alpha(\alpha - 1)x_1^{\alpha-2}x_2^{1-\alpha}, \quad H_{22} = -\alpha(1-\alpha)x_1^{\alpha}x_2^{-\alpha-1}, \quad H_{12} = \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha}.$$

- Second leading principal minor:

$$\Delta_2 = H_{11}H_{22} - H_{12}^2 = -2\alpha^2(1-\alpha)^2 x_1^{2\alpha-2}x_2^{-2\alpha} < 0.$$

- For 2×2 NSD: need $H_{11} \leq 0$, $H_{22} \leq 0$, and $\Delta_2 \geq 0$ (Section 4.2)
- $\Delta_2 < 0$, so H_u is not NSD — u is not concave

Yet u is quasiconcave (previous exercise), so quasiconcavity is genuinely **weaker** than concavity.

4.2 Exercise: Testing quasiconcavity Take home

Exercise. Is $f(x, y) = xy$ quasiconcave on $\mathbb{R}^2_{++}$?

Solution. Show $S_t = \{(x, y) \in \mathbb{R}^2_{++} : xy \geq t\}$ is convex for every t.

- If $t \leq 0$: $S_t = \mathbb{R}^2_{++}$, convex
- If $t > 0$: $S_t = \{(x, y) : g(x, y) \geq \ln t\}$ with $g(x, y) = \ln x + \ln y$
- g is concave: $H_g = \mathrm{diag}(-1/x^2, -1/y^2)$ is negative definite
- $\{(x, y) : g(x, y) \geq \ln t\}$ is an **upper level set** of g; recall: upper level sets of a concave function are convex
- So S_t is convex

Therefore f is quasiconcave.

 Take home

Exercise. For CES utility $u(x_1, x_2) = (x_1^\rho + x_2^\rho)^{1/\rho}$ on $\mathbb{R}^2_{++}$ $(\rho \neq 0)$:

(a) Show u is quasiconcave when $\rho \leq 1$.

(b) Show u is not quasiconcave when $\rho > 1$.

Recall. Quasiconcavity is preserved under strictly increasing transforms (Section 4.2).

4.2 Exercise: CES quasiconcavity (solution) Take home

(a) Let $g(x_1, x_2) = x_1^\rho + x_2^\rho$ and $\varphi(t) = t^{1/\rho}$.

- For $0 < \rho \leq 1$: $H_g = \rho(\rho - 1)\,\mathrm{diag}(x_1^{\rho-2}, x_2^{\rho-2})$ is NSD, so g is concave
- Concave $\Rightarrow$ quasiconcave; φ strictly increasing, so $u = \varphi \circ g$ is quasiconcave (recall above)
- If $\rho = 1$: $u = x_1 + x_2$ is linear

4.2 Exercise: CES quasiconcavity (solution, b) Take home

(b) Fix $\rho > 1$. **Goal:** show $u(\lambda \mathbf{x} + (1-\lambda)\mathbf{y}) \geq \min\{u(\mathbf{x}), u(\mathbf{y})\}$ does **not** hold for some $\mathbf{x}, \mathbf{y}, \lambda$ (violating segment quasiconcavity).

Take $\varepsilon \in (0,1)$, $\mathbf{x} = (1, \varepsilon)$, $\mathbf{y} = (\varepsilon, 1)$, $\lambda = \frac{1}{2}$.

- By symmetry: $u(\mathbf{x}) = u(\mathbf{y}) = \min\{u(\mathbf{x}), u(\mathbf{y})\} = (1+\varepsilon^\rho)^{1/\rho}$
- At $\lambda = \frac{1}{2}$: $u(\lambda \mathbf{x} + (1-\lambda)\mathbf{y}) = \left(2\left(\frac{1+\varepsilon}{2}\right)^\rho\right)^{1/\rho} = (1+\varepsilon)\, 2^{1/\rho - 1}$
- Let $h(\varepsilon) := (1+\varepsilon^\rho)^{1/\rho} - (1+\varepsilon)2^{1/\rho-1}$. For $\rho > 1$: $2^{1/\rho-1} < 1$, so $h(0) > 0$
- By continuity, pick $\varepsilon > 0$ small with $h(\varepsilon) > 0$, i.e. $u(\lambda \mathbf{x} + (1-\lambda)\mathbf{y}) < \min\{u(\mathbf{x}), u(\mathbf{y})\}$ — not quasiconcave

4.2 Why quasiconcavity matters

Theorem. If f is quasiconcave and the feasible set $\mathcal{B}$ is convex, then any feasible point with no first-order improving direction is a global maximum.

Intuition:

- Quasiconcavity ensures "no valleys" in the upper direction
- With convex feasible set, any local peak is the global peak

Cobb–Douglas is quasiconcave, so the FOC candidate gives the global max on a convex budget set.

4.3 Constrained optimization problem

Problem. Maximize $f(\mathbf{x})$ subject to $h(\mathbf{x}) = 0$.

The constraint defines a curve (or surface) in the domain.

We seek the highest point of f on this surface.

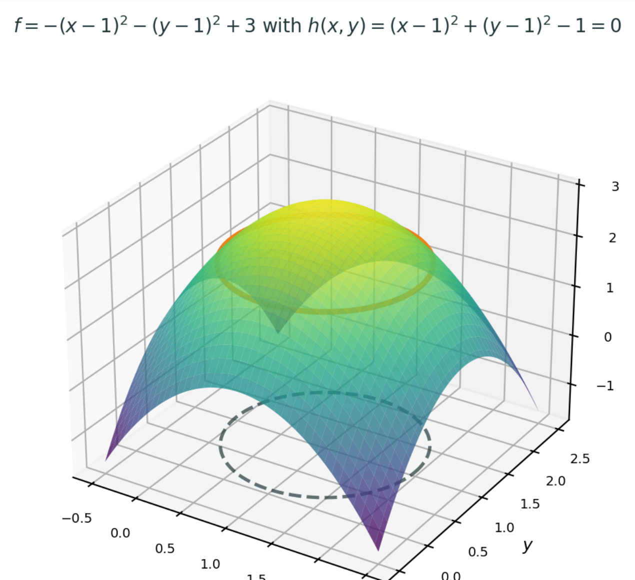
Key insight: We cannot simply set $\nabla f = 0$ — must respect the constraint.

4.3 Example: Utility maximization

Example. Maximize utility $u(x_1, x_2)$ subject to budget $p_1x_1 + p_2x_2 = w$.

The budget line is the constraint surface in (x_1, x_2) space.

The optimum is the highest indifference curve that still touches the budget line.



$f = -(x-1)^2 - (y-1)^2 + 3$ with $h(x, y) = (x-1)^2 + (y-1)^2 - 1 = 0$
3
2
1
0
−1
−0.5
0.0
0.5
1.0
2.5
2.0
1.5
1.0
0.5
y

4.3 The tangent argument

At a constrained optimum $\mathbf{x}^*$ on the constraint surface:

- Any feasible direction $\mathbf{d}$ (tangent to constraint) satisfies $\nabla h(\mathbf{x}^*) \cdot \mathbf{d} = 0$
- If $\nabla f(\mathbf{x}^*) \cdot \mathbf{d} \neq 0$, we could move along constraint to increase f
- At optimum, no feasible improving direction exists

Conclusion. $\nabla f(\mathbf{x}^*)$ must be parallel to $\nabla h(\mathbf{x}^*)$.

That is: $\nabla f(\mathbf{x}^*) = -\lambda \nabla h(\mathbf{x}^*)$ for some scalar λ.

4.3 Lagrangian and first-order conditions

Theorem. At a constrained optimum (with regularity), there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(\mathbf{x}^*) + \lambda \nabla h(\mathbf{x}^*) = \mathbf{0}, \quad h(\mathbf{x}^*) = 0.$$

Lagrangian. $\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda h(\mathbf{x})$.

- Stationarity: $\nabla_{\mathbf{x}} \mathcal{L} = \nabla f + \lambda \nabla h = \mathbf{0}$
- Constraint: $\partial \mathcal{L} / \partial \lambda = h = 0$

λ is the **Lagrange multiplier**.

4.3 Sensitivity without the Lagrangian

Problem: $\max_x f(x)$ subject to $h(x) = c$. At an optimum, $\nabla f(x^*) = \lambda \nabla h(x^*)$.

Relax the constraint a little: move to $x^* + dx$ with

$$h(x^* + dx) = c + dc.$$

First-order expansion: $\nabla h \cdot dx = dc$. Total differential of the objective: $df = \nabla f \cdot dx$.

Substitute the FOC:

$$df = \nabla f \cdot dx = \lambda \nabla h \cdot dx = \lambda \, dc.$$

Hence $\dfrac{dV}{dc} = \lambda$: the multiplier is the sensitivity of f to relaxing the constraint.

4.3 Connect to the Lagrangian

Same problem: $\max_x f(x)$ s.t. $h(x) = c$, with $\mathcal{L}(x, \lambda; c) = f(x) + \lambda\big(c - h(x)\big)$.

Let $V(c) = f(x^*(c))$. Binding constraint $\Rightarrow$ $V(c) = \mathcal{L}(x^*(c), \lambda^*(c); c)$.

Chain rule: FOCs kill $\nabla_x \mathcal{L} \cdot dx^*/dc$ and $(\partial\mathcal{L}/\partial\lambda)\, d\lambda^*/dc$, so

$$\frac{dV}{dc} = \frac{\partial \mathcal{L}}{\partial c}.$$

And $\partial\mathcal{L}/\partial c = \lambda^*$ by inspection of $\mathcal{L}$.

Summary. $\dfrac{dV}{dc} = \dfrac{\partial \mathcal{L}}{\partial c} = \lambda^*$.

4.3 Exercise: Cobb–Douglas with budget Together

Exercise. Maximize $u(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha}$ subject to $p_1 x_1 + p_2 x_2 = w$.

Solution.

- Lagrangian: $\mathcal{L} = u + \lambda(w - p_1 x_1 - p_2 x_2)$
- FOCs: $\frac{\partial u}{\partial x_1} = \lambda p_1$, $\frac{\partial u}{\partial x_2} = \lambda p_2$
- Expanding: $\frac{\alpha u}{x_1} = \lambda p_1$, $\frac{(1-\alpha) u}{x_2} = \lambda p_2$
- Taking ratio: $\frac{\alpha x_2}{(1-\alpha) x_1} = \frac{p_1}{p_2}$

With budget: $p_1 x_1 = \alpha w$, $p_2 x_2 = (1-\alpha) w$ (constant expenditure shares, Day 1).

From the FOC, $\lambda = \frac{\partial u / \partial x_1}{p_1}$: λ is the utility gain from spending one more dollar on x_1 (same for x_2 at the optimum).

Exercise. In the Cobb–Douglas solution, compute λ^* and interpret it.

Solution. From FOC and $p_1 x_1 = \alpha w$: $\lambda = u/w$. Plug in demands:

$$\lambda^* = \frac{\alpha^\alpha (1-\alpha)^{1-\alpha}}{p_1^\alpha p_2^{1-\alpha}}.$$

In general we only know $\lambda^* = dV/dw$ (shadow price). Here something stronger holds: $\lambda^* = u^*/w = V/w$.

That is CD-specific: MUI equals *average* utility per dollar, not just the marginal.

Since the explicit λ^* does not depend on w, integrate $dV/dw = \lambda^*$ with $V(0) = 0$ to get $V(w) = \lambda^* w$, which is consistent with $\lambda^* = V/w$.

4.3 Why is MUI constant in w?

At an optimum, $MRS = p_1/p_2$. For CD, $MRS = \frac{\alpha}{1-\alpha}\frac{x_2}{x_1}$ depends only on the consumption *ratio*.

So x_1/x_2 is pinned down by prices alone — it does not change when w rises.

Key intuition. The mix is fixed; if wealth increases, consumption of each good rises *proportionally*:

$$x^*(w) = w\,x^*(1).$$

CD is HD1, so $V(w) = u(w\,x^*(1)) = w\,V(1)$, hence $dV/dw = V(1)$ is constant in w.

4.3 Exercise: Cost minimization   Take home

Exercise. Minimize $C(q_1, q_2) = r_1 q_1 + r_2 q_2$ subject to $F(q_1, q_2) = \bar{y}$.

Solution.

- Lagrangian: $\mathcal{L} = r_1 q_1 + r_2 q_2 + \lambda(\bar{y} - F(q_1, q_2))$
- FOCs: $r_1 = \lambda F_1$, $r_2 = \lambda F_2$ where $F_i = \frac{\partial F}{\partial q_i}$
- Therefore: $\frac{r_1}{r_2} = \frac{F_1}{F_2}$

The input price ratio equals the marginal rate of technical substitution.

Same logic: $\lambda = dC^*/d\bar{y}$, the marginal cost of relaxing the output constraint.

4.4 Value functions and parameters

Economic problems depend on parameters: prices, income, technology.

Value function. $V(\theta) = \max_{\mathbf{x}} f(\mathbf{x}; \theta)$ subject to constraints.

We want to know: how does the optimal value change when θ shifts?

Key question: Do we need to account for how $\mathbf{x}^*(\theta)$ changes?

4.4 Unconstrained envelope theorem

Theorem. For $V(\theta) = \max_{\mathbf{x}} f(\mathbf{x}; \theta)$ with optimizer $\mathbf{x}^*(\theta)$:

$$\frac{dV}{d\theta} = \frac{\partial f}{\partial \theta}\bigg|_{\mathbf{x}=\mathbf{x}^*(\theta)}.$$

Proof. By chain rule:

$$\frac{dV}{d\theta} = \underbrace{\nabla_{\mathbf{x}} f \cdot \frac{d\mathbf{x}^*}{d\theta}}_{=0 \text{ by FOC}} + \frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial \theta}.$$

Intuition. First-order gains from adjusting $\mathbf{x}$ vanish at optimum. Only direct effects of θ on f matter.

4.4 Constrained envelope theorem

With constraints and Lagrangian $\mathcal{L} = f + \sum \lambda_k h_k$:

$$\frac{dV}{d\theta} = \frac{\partial \mathcal{L}}{\partial \theta}\bigg|_{(\mathbf{x}^*, \boldsymbol{\lambda}^*)}.$$

Proof. Differentiate $V(\theta) = \mathcal{L}(\mathbf{x}^*(\theta), \boldsymbol{\lambda}^*(\theta); \theta)$:

- Terms with $\partial \mathbf{x}^* / \partial \theta$ vanish by stationarity ($\nabla_{\mathbf{x}} \mathcal{L} = 0$)
- Terms with $\partial \lambda_k / \partial \theta$ vanish by constraints ($h_k = 0$)

Shadow prices absorb the effect of constraint relaxation.

Exercise. For $\pi^*(p) = \max_{q \geq 0} pq - C(q)$, find $\frac{d\pi^*}{dp}$.

Solution by envelope.

- $\frac{d\pi^*}{dp} = \frac{\partial(pq - C(q))}{\partial p}\Big|_{q^*} = q^*(p)$
- The supply function!

Verify explicitly.

- FOC: $p = C'(q^*)$ defines $q^*(p)$
- $\pi^*(p) = pq^*(p) - C(q^*(p))$
- $\frac{d\pi^*}{dp} = q^* + p\frac{dq^*}{dp} - C'(q^*)\frac{dq^*}{dp} = q^* + \underbrace{(p - C'(q^*))}_{=0}\frac{dq^*}{dp} = q^*$

Envelope gives this directly without computing $\frac{dq^*}{dp}$.

4.4 Budget set and indirect utility

Budget set. $B(p, w) = \{\mathbf{x} \geq \mathbf{0} : \; p \cdot \mathbf{x} \leq w\}$.

Consumer problem. $\max_{\mathbf{x}} u(\mathbf{x})$ subject to $\mathbf{x} \in B(p, w)$.

Let $\mathbf{x}^*(p, w)$ be the Marshallian demand (optimizer).

Indirect utility. The value function of this problem:

$$V(p, w) = \max_{\mathbf{x} \in B(p,w)} u(\mathbf{x}) = u\big(\mathbf{x}^*(p, w)\big).$$

V is maximized utility as a function of prices and wealth — not of the goods themselves.

4.4 Exercise: Roy's identity Together

Exercise. Show that

$$\frac{\partial V/\partial p_i}{\partial V/\partial w} = -x_i^*(p, w).$$

Solution. Envelope: $\partial V/\partial p_i = -\lambda^* x_i^*$, $\partial V/\partial w = \lambda^*$. Ratio:

$$\frac{-\lambda^* x_i^*}{\lambda^*} = -x_i^*.$$

So Marshallian demand is recovered from V without re-solving the consumer problem.

4.4 How p_i changes V: two channels

Why is $\partial V/\partial p_i = -\lambda^* x_i^*$? Write V via $\mathcal{L} = u(\mathbf{x}) + \lambda(w - p \cdot \mathbf{x})$, so $V = \mathcal{L}\big(\mathbf{x}^*, \lambda^*; p, w\big)$.

Differentiate in p_i:

$$\frac{\partial V}{\partial p_i} = \underbrace{\nabla_{\mathbf{x}}\mathcal{L} \cdot \frac{\partial \mathbf{x}^*}{\partial p_i}}_{=0 \text{ (FOC)}} + \underbrace{\frac{\partial \mathcal{L}}{\partial \lambda}\frac{\partial \lambda^*}{\partial p_i}}_{=0 \text{ (budget binds)}} + \underbrace{\frac{\partial \mathcal{L}}{\partial p_i}}_{=-\lambda^* x_i^*}.$$

Two channels.

- Fix $\mathbf{x}$: higher p_i costs x_i^* more dollars for the same basket — valued at λ^* utils each.
- Reoptimize $\mathbf{x}^*$ (and λ^*): first-order effect is zero at an optimum.

4.5 Problems with inequality constraints

Problem. Maximize $f(\mathbf{x})$ subject to $g_j(\mathbf{x}) \leq 0$ for $j = 1, \ldots, m$.

Binding vs. slack constraints.

- **Binding (active):** $g_j(\mathbf{x}^*) = 0$ — constraint holds with equality
- **Slack (inactive):** $g_j(\mathbf{x}^*) < 0$ — constraint not tight

Only binding constraints matter for local optimality.

Slack constraints can be ignored locally.

4.5 Feasible directions and the cone

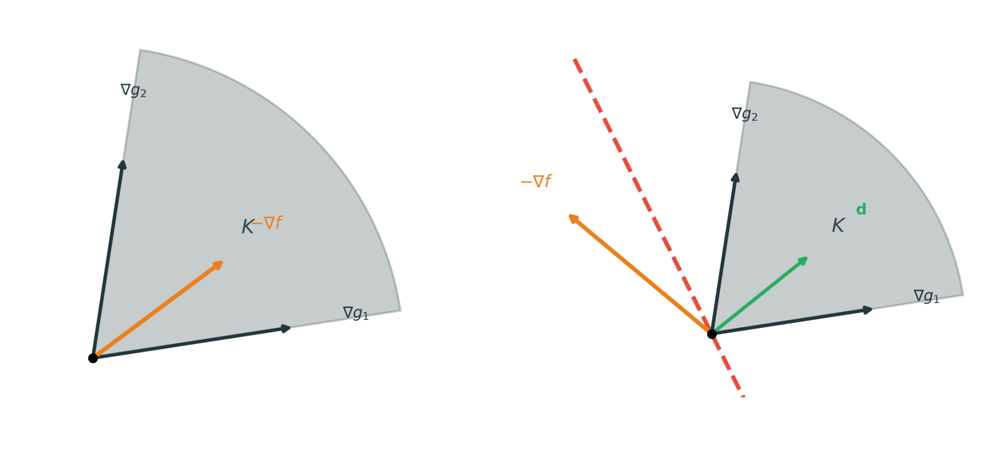
At a boundary point with active constraints, **feasible directions** form a convex cone:

$$D = \{\mathbf{d} : \nabla g_j(\mathbf{x}^*) \cdot \mathbf{d} \leq 0 \text{ for all active } j\}.$$

A direction $\mathbf{d}$ is **improving** if $\nabla f \cdot \mathbf{d} > 0$ (increases objective).

For a local maximum: no feasible direction can be improving.

4.5 Geometric intuition of Farkas lemma



4.5 Statement of Farkas lemma

Farkas Lemma (purely geometric).

Let K be a closed convex cone and $\mathbf{y}$ a vector. Either:

1. $\mathbf{y} \in K$: the vector lies in the cone
2. $\mathbf{y} \notin K$: there exists a separating hyperplane with normal $\mathbf{d}$ such that
    - $\mathbf{k} \cdot \mathbf{d} \leq 0$ for all $\mathbf{k} \in K$ (cone on one side)
    - $\mathbf{y} \cdot \mathbf{d} > 0$ (vector on the other side)

The hyperplane strictly separates the cone from the vector.

4.5 Applying Farkas to optimization

Maximize f subject to $g_j \leq 0$. Let $K = \mathrm{cone}\{\nabla g_j : g_j \text{ active}\}$.

Farkas, applied: either $\nabla f \in K$, or there is a feasible improving direction.

Case 1 (optimal). $\nabla f \in K$: ∇f is a nonnegative combination of the active ∇g_j. With $\mathcal{L} = f + \sum_j \lambda_j g_j$, this is exactly

$$\nabla f + \sum_j \lambda_j \nabla g_j = \mathbf{0} \quad \text{for some } \lambda_j \leq 0.$$

(Why ≤ 0? Stationarity gives $\nabla f = -\sum \lambda_j \nabla g_j$; to lie in K we need $-\lambda_j \geq 0$.)

Case 2 (not optimal). $\nabla f \notin K$: some $\mathbf{d}$ with $\nabla g_j \cdot \mathbf{d} \leq 0$ (feasible) and $\nabla f \cdot \mathbf{d} > 0$ (improving).

4.5 Inequality problem with multipliers

At an optimum, Case 2 is ruled out — so Case 1 holds:

$$\nabla f(\mathbf{x}^*) + \sum_{\text{active}} \lambda_j \nabla g_j(\mathbf{x}^*) = \mathbf{0}, \quad \lambda_j \le 0.$$

Binding: $\lambda_j < 0$ possible (these constraints bite).

Slack: $\lambda_j = 0$ (inactive constraints drop out of $\mathcal{L}$).

Same λ as in the Lagrangian — no extra symbols.

4.5 KKT conditions

Theorem (KKT necessary conditions). At a local maximum (with constraint qualification), there exist multipliers λ_j such that:

1. **Stationarity:** $\nabla f(\mathbf{x}^*) + \sum_j \lambda_j \nabla g_j(\mathbf{x}^*) = \mathbf{0}$
2. **Primal feasibility:** $g_j(\mathbf{x}^*) \leq 0$ for all j
3. **Dual feasibility:** $\lambda_j \leq 0$ (for $g_j \leq 0$, max problem, $\mathcal{L} = f + \sum \lambda_j g_j$)
4. **Complementary slackness:** $\lambda_j g_j(\mathbf{x}^*) = 0$

CS says: multiplier is zero for slack constraints ($g_j < 0 \Rightarrow \lambda_j = 0$).

Stationarity says: ∇f is a nonnegative combination of active ∇g_j (since $\lambda_j \leq 0$).

4.5 Sign convention summary

With $\mathcal{L} = f + \sum_j \lambda_j g_j$ (always $+$):

Problem	Constraint	λ sign
$\max f$	$g \leq 0$	$\lambda \leq 0$
$\max f$	$g \geq 0$	$\lambda \geq 0$
$\min f$	$g \leq 0$	$\lambda \geq 0$
$\min f$	$g \geq 0$	$\lambda \leq 0$

Memory rule (multiply signs). Assign: max $= +$, min $= -$; $\leq = -$; $\geq = +$. The sign of λ is the product:

$$\underbrace{\max/\min}_{\pm} \times \underbrace{g \ \leq / \geq}_{\pm} \ = \ \mathrm{sign}(\lambda).$$

Example: max and $g \leq 0 \Rightarrow (+) \times (-) = - \Rightarrow \lambda \leq 0$.

4.5 Exercise: Budget with nonnegativity Together

Exercise. Maximize $u(x_1, x_2) = x_1 x_2$ subject to $x_1 + x_2 \leq 1$, $x_1 \geq 0$, $x_2 \geq 0$.

$g_1 = x_1 + x_2 - 1 \leq 0$, $g_2 = -x_1 \leq 0$, $g_3 = -x_2 \leq 0$, $\lambda_j \leq 0$.

$$\mathcal{L} = x_1 x_2 + \lambda_1(x_1 + x_2 - 1) + \lambda_2(-x_1) + \lambda_3(-x_2).$$

Stationarity: $x_2 + \lambda_1 - \lambda_2 = 0, \quad x_1 + \lambda_1 - \lambda_3 = 0.$
CS: $\lambda_1(x_1 + x_2 - 1) = 0, \quad \lambda_2 x_1 = 0, \quad \lambda_3 x_2 = 0.$

Method. In general each g_j binds or not $\Rightarrow 2^3 = 8$ patterns. We illustrate three: interior max, a corner that fails KKT, and a point where KKT holds but is not a max.

Stationarity: $x_2 + \lambda_1 - \lambda_2 = 0, \quad x_1 + \lambda_1 - \lambda_3 = 0.$
CS: $\lambda_1(x_1 + x_2 - 1) = 0, \quad \lambda_2 x_1 = 0, \quad \lambda_3 x_2 = 0 \quad (\lambda_j \leq 0).$

Case. $x_1 + x_2 = 1$ and $x_1, x_2 > 0 \Rightarrow \lambda_2 = \lambda_3 = 0.$

- Stationarity: $x_2 + \lambda_1 = 0$, $x_1 + \lambda_1 = 0 \Rightarrow x_1 = x_2 = -\lambda_1$
- Budget: $x_1 + x_2 = 1 \Rightarrow x_1 = x_2 = \frac{1}{2}$, $\lambda_1 = -\frac{1}{2} \leq 0$

KKT holds with $u = \frac{1}{4}$. This is the maximizer.

Stationarity: $x_2 + \lambda_1 - \lambda_2 = 0, \quad x_1 + \lambda_1 - \lambda_3 = 0.$
CS: $\lambda_1(x_1 + x_2 - 1) = 0, \quad \lambda_2 x_1 = 0, \quad \lambda_3 x_2 = 0 \quad (\lambda_j \le 0).$

Case. Budget binds and $x_1 = 0$, x_2 slack: $(0, 1)$. Then g_3 slack $\Rightarrow \lambda_3 = 0$.

- Stationarity in x_2: $\lambda_1 = 0$
- Stationarity in x_1: $1 - \lambda_2 = 0 \Rightarrow \lambda_2 = 1 > 0$

Violates $\lambda_2 \le 0 \Rightarrow$ **not KKT** $\Rightarrow (0, 1)$ cannot be optimal. (Same for $(1, 0)$ by symmetry.)

4.5 Case: Both axes bind, KKT but not max

Together

Stationarity: $x_2 + \lambda_1 - \lambda_2 = 0, \quad x_1 + \lambda_1 - \lambda_3 = 0.$
CS: $\lambda_1(x_1 + x_2 - 1) = 0, \quad \lambda_2 x_1 = 0, \quad \lambda_3 x_2 = 0 \quad (\lambda_j \leq 0).$

Case. $x_1 = x_2 = 0$ (both nonnegativity bind), budget slack.

CS $\Rightarrow \lambda_1 = 0$. Stationarity $\Rightarrow \lambda_2 = \lambda_3 = 0$.

KKT holds, but $u(0,0) = 0 < \frac{1}{4} = u(\frac{1}{2}, \frac{1}{2})$.

So KKT is necessary (under CQ), not sufficient: must still compare objective values.

4.5 Exercise: Cost minimization Together

Exercise. A firm chooses inputs $z = (z_1, z_2)$ to produce at least $\bar{z}$ units with technology $f(z) = z_1 + z_2$ (perfect substitutes). Input prices are $p = (p_1, p_2) > 0$.

$$\min_{z \geq 0} \; p \cdot z \quad \text{s.t.} \quad z_1 + z_2 \geq \bar{z}.$$

Setup. Write $g_1 = \bar{z} - z_1 - z_2 \leq 0$, $g_2 = -z_1 \leq 0$, $g_3 = -z_2 \leq 0$. For a *min* problem with $g_j \leq 0$: $\lambda_j \geq 0$.

$$\mathcal{L} = p_1 z_1 + p_2 z_2 + \lambda_1(\bar{z} - z_1 - z_2) + \lambda_2(-z_1) + \lambda_3(-z_2).$$

Stationarity: $p_1 - \lambda_1 - \lambda_2 = 0, \quad p_2 - \lambda_1 - \lambda_3 = 0.$
CS: $\lambda_1(\bar{z} - z_1 - z_2) = 0, \quad \lambda_2 z_1 = 0, \quad \lambda_3 z_2 = 0.$

Stationarity: $p_1 = \lambda_1 + \lambda_2, \quad p_2 = \lambda_1 + \lambda_3.$
CS: $\lambda_1(\bar{z} - z_1 - z_2) = 0, \quad \lambda_2 z_1 = 0, \quad \lambda_3 z_2 = 0 \quad (\lambda_j \geq 0).$

Interior in both inputs $(z_1, z_2 > 0)$. Then $\lambda_2 = \lambda_3 = 0$, so $p_1 = p_2 = \lambda_1$. Output binds: $z_1 + z_2 = \bar{z}$. Any split works iff $p_1 = p_2$.

If $p_1 < p_2$. Cannot have $\lambda_2 = \lambda_3 = 0$. Check the cases with a positive multiplier:

- $\lambda_2 > 0$: CS $\Rightarrow z_1 = 0$, so $z_2 = \bar{z}$. Then $\lambda_3 = 0$, $\lambda_1 = p_2$, $\lambda_2 = p_1 - p_2 < 0$. Violates $\lambda_2 \geq 0$. **Fails**.
- $\lambda_3 > 0$: CS $\Rightarrow z_2 = 0$, so $z_1 = \bar{z}$. Then $\lambda_2 = 0$, $\lambda_1 = p_1$, $\lambda_3 = p_2 - p_1 > 0$. **Works** (use only the cheaper input).

4.6 Key intuition: one-sided vs. two-sided wall

Inequality $g(\mathbf{x}) \le 0$: a **one-sided wall**. Feasible moves: $\nabla g \cdot \mathbf{d} \le 0$. For a max with one active g and $\mathcal{L} = f + \lambda g$:

1. **Binding + stationarity** $\Rightarrow$ ∇f on the *same line* as ∇g: $\nabla f = -\lambda \nabla g$ (same or opposite).
2. Binding needs $\nabla f \cdot \nabla g > 0 \Rightarrow$ same way $\Rightarrow \lambda < 0$.
3. If $\lambda > 0$, λg rewards raising g (violate $g \le 0$). $\lambda < 0$ holds that incentive back on the bound.

Key difference. When binding: inequality — you know which way raises f (along ∇g, into the forbidden side); equality — you do not (with or against ∇h may raise or lower f). So equality λ is unrestricted.

4.6 The combined Lagrangian

For problems with both equalities and inequalities:

$$\mathcal{L} = f + \underbrace{\sum_k \lambda_k h_k}_{\text{equality (two-sided)}} + \underbrace{\sum_j \lambda_j g_j}_{\text{inequality (one-sided)}} .$$

Sign convention (max problem):

Multiplier	Sign rule
Equality λ_k	unrestricted (two-sided wall)
Inequality λ_j	$\lambda_j \leq 0$ if $g_j \leq 0$; $\lambda_j \geq 0$ if $g_j \geq 0$

Short rule. Sign of $\lambda = (\max + / \min -) \times (g \geq 0 + / g \leq 0 -)$.

Appendix for interest only

Additional proofs and extensions (same order as main text):

1. A.1 Unconstrained optimization: FOC and SOC (1D)
2. A.2 Concavity: (1) $\Leftrightarrow$ (2)
3. A.3 Concavity: (2) $\Leftrightarrow$ (3)
4. A.4 Proof: multivariate (2) $\Leftrightarrow$ (3)
5. A.5 Multivariate SOC
6. A.6 Log-concavity
7. A.7 Additional exercises

A.1 Proof: First-Order Condition (1D)

for interest only

Theorem. If x^* is an interior local maximum and f is differentiable at x^*, then $f'(x^*) = 0$.

Proof.

- If $f'(x^*) > 0$: then $f(x^* + h) > f(x^*)$ for small $h > 0$, contradiction
- If $f'(x^*) < 0$: then $f(x^* + h) > f(x^*)$ for small $h < 0$, contradiction

A.1 Proof: Necessary SOC (1D)

for interest only

Theorem. If x^* is an interior local maximum and $f \in C^2$, then $f''(x^*) \leq 0$.

Proof. Suppose $f''(x^*) > 0$. Taylor at x^*:

$$f(x^* + h) = f(x^*) + \frac{1}{2} f''(x^*) h^2 + o(h^2).$$

- For small $h \neq 0$, the quadratic term dominates the remainder
- Since $f''(x^*) > 0$, we get $f(x^* + h) > f(x^*)$ — contradiction
- Therefore $f''(x^*) \leq 0$

A.1 Proof: Sufficient SOC (1D)

for interest only

Theorem. If $f'(x^*) = 0$ and $f''(x^*) < 0$, then x^* is a strict local maximum.

Proof. Taylor expansion around x^*:

$$f(x^* + h) = f(x^*) + \underbrace{f'(x^*)}_{=0} h + \tfrac{1}{2} f''(x^*) h^2 + o(h^2).$$

- For small h, the h^2 term dominates the remainder
- Since $f''(x^*) < 0$, we have $f(x^* + h) < f(x^*)$ for small $h \neq 0$

A.2 Proof (1) $\Rightarrow$ (2)

for interest only

Proof. Fix x and $y > x$. Set $h = y - x > 0$.

For $t \in (0, 1]$, the point on the segment is

$$x + th = (1 - t)x + ty.$$

- By concavity: $f(x + th) \geq (1 - t)f(x) + tf(y)$
- Rearrange: $f(x + th) - f(x) \geq t(f(y) - f(x))$
- Divide by $th > 0$:
$$\frac{f(x + th) - f(x)}{th} \geq \frac{f(y) - f(x)}{y - x}.$$
- Let $t \to 0^+$: $f'(x) \geq \dfrac{f(y) - f(x)}{y - x}$
- So $f(y) \leq f(x) + f'(x)(y - x)$. The case $y < x$ is similar

A.2 Proof (2) $\Rightarrow$ (1)

for interest only

Proof. Fix x, y, and $\lambda \in [0,1]$. Let $u = \lambda x + (1-\lambda)y$.

- Tangent at u: $f(x) \leq f(u) + f'(u)(x-u)$ and $f(y) \leq f(u) + f'(u)(y-u)$
- Weight by $(1-\lambda)$ and λ, then add:

$$(1-\lambda)f(x) + \lambda f(y) \leq f(u) + f'(u)\big[(1-\lambda)(x-u) + \lambda(y-u)\big].$$

- The bracket is 0 because $u = \lambda x + (1-\lambda)y$
- So $(1-\lambda)f(x) + \lambda f(y) \leq f(u) = f(\lambda x + (1-\lambda)y)$

A.3 Proof (2) $\Rightarrow$ (3)

for interest only

Proof. Since (1) $\Leftrightarrow$ (2), use the chord form. From concavity with endpoints $x+h$, $x-h$ and $\lambda = \frac{1}{2}$:

$$f(x) = f\left(\tfrac{1}{2}(x+h) + \tfrac{1}{2}(x-h)\right) \geq \tfrac{1}{2}f(x+h) + \tfrac{1}{2}f(x-h).$$

So $f(x+h) + f(x-h) \leq 2f(x)$ for all $h > 0$.

Rearrange and divide by $h > 0$:

$$\frac{f(x+h) - f(x)}{h} \leq \frac{f(x) - f(x-h)}{h}.$$

Bring to one side and divide by h again:

$$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} \leq 0.$$

Let $h \to 0$: limit is $f''(x) \leq 0$.

Rigorous version: next slides

A.3 Proof (3) $\Rightarrow$ (2)

for interest only

Proof. Fix any x and y. Taylor around x:

$$f(y) = f(x) + f'(x)(y - x) + \frac{1}{2} f''(\xi)(y - x)^2,$$

where ξ lies between x and y.

- Since $f''(\xi) \leq 0$ and $(y - x)^2 \geq 0$, the second-order term is ≤ 0
- Therefore: $f(y) \leq f(x) + f'(x)(y - x)$
- This is the tangent characterization of concavity

A.3 Rigorous proof: concave $\Rightarrow f'' \leq 0$ (I)

for interest only

Theorem. If f is concave and $f \in C^2$, then $f''(x) \leq 0$ for all x.

Rigorous proof using Taylor expansion with remainder.

From concavity with endpoints $x+h$, $x-h$ and $\lambda = \frac{1}{2}$:

$$f(x) \geq \tfrac{1}{2} f(x+h) + \tfrac{1}{2} f(x-h) \quad \Leftrightarrow \quad f(x+h) + f(x-h) \leq 2f(x).$$

Taylor expansions around x:

$$f(x+h) = f(x) + f'(x)h + \frac{1}{2} f''(x)h^2 + \frac{f'''(\xi_1)}{6} h^3$$
$$f(x-h) = f(x) - f'(x)h + \frac{1}{2} f''(x)h^2 - \frac{f'''(\xi_2)}{6} h^3$$

where ξ_1 lies between x and $x+h$, and ξ_2 between $x-h$ and x.

A.3 Rigorous proof: concave $\Rightarrow f'' \leq 0$ (II)

for interest only

Adding the two Taylor expansions:

$$f(x+h) + f(x-h) = 2f(x) + f''(x)h^2 + \frac{f'''(\xi_1) - f'''(\xi_2)}{6}h^3.$$

- By concavity: $f''(x)h^2 + \frac{f'''(\xi_1)-f'''(\xi_2)}{6}h^3 \leq 0$
- Divide by $h^2 > 0$:
$$f''(x) + \frac{f'''(\xi_1) - f'''(\xi_2)}{6}h \leq 0$$
- Let $h \to 0$: third-order terms vanish, leaving $f''(x) \leq 0$

A.4 Proof (3) $\Rightarrow$ (2)

for interest only

Proof of (2) $\Leftrightarrow$ (3) for multivariate concavity.

Proof. Taylor around $\mathbf{x}$:

$$f(\mathbf{x}+\mathbf{h}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{h} + \frac{1}{2}\mathbf{h}^\top H_f(\xi)\mathbf{h},$$

where ξ lies on the segment from $\mathbf{x}$ to $\mathbf{x}+\mathbf{h}$.

- If $H_f(\xi)$ is NSD, then $\mathbf{h}^\top H_f(\xi)\mathbf{h} \leq 0$
- So $f(\mathbf{x}+\mathbf{h}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{h}$
- Writing $\mathbf{y} = \mathbf{x}+\mathbf{h}$ gives the tangent characterization

A.4 Proof (2) $\Rightarrow$ (3)

for interest only

Proof. Fix $\mathbf{x}$ and any $\mathbf{h} \in \mathbb{R}^n$. By concavity with endpoints $\mathbf{x} + \mathbf{h}$, $\mathbf{x} - \mathbf{h}$ and $\lambda = \frac{1}{2}$:

$$f(\mathbf{x}) \geq \tfrac{1}{2} f(\mathbf{x} + \mathbf{h}) + \tfrac{1}{2} f(\mathbf{x} - \mathbf{h}).$$

- Taylor expand $f(\mathbf{x} \pm \mathbf{h})$ around $\mathbf{x}$ and add
- The first-order terms cancel; the remainder gives

$$\mathbf{h}^\top H_f(\mathbf{x})\mathbf{h} + o(\|\mathbf{h}\|^2) \leq 0$$

- Let $\|\mathbf{h}\| \to 0$: $\mathbf{h}^\top H_f(\mathbf{x})\mathbf{h} \leq 0$ for all $\mathbf{h}$
- So $H_f(\mathbf{x})$ is NSD

A.5 Proof: Necessary SOC (multivariate)

for interest only

Theorem. If $\mathbf{x}^*$ is an interior local maximum and $f \in C^2$, then $H_f(\mathbf{x}^*)$ is negative semidefinite.

Proof. Suppose some eigenvalue $\lambda > 0$ with unit eigenvector $\mathbf{v}$ ($H_f(\mathbf{x}^*)\mathbf{v} = \lambda\mathbf{v}$, $\|\mathbf{v}\| = 1$).
Taylor at $\mathbf{x}^*$:

$$f(\mathbf{x}^* + t\mathbf{v}) = f(\mathbf{x}^*) + \tfrac{1}{2}t^2\mathbf{v}^\top H_f(\mathbf{x}^*)\mathbf{v} + o(t^2).$$

- Since $\mathbf{v}$ is an eigenvector: $\mathbf{v}^\top H_f(\mathbf{x}^*)\mathbf{v} = \mathbf{v}^\top(\lambda\mathbf{v}) = \lambda\,\mathbf{v}^\top\mathbf{v} = \lambda\|\mathbf{v}\|^2 = \lambda$
- So $f(\mathbf{x}^* + t\mathbf{v}) = f(\mathbf{x}^*) + \tfrac{1}{2}t^2\lambda + o(t^2)$
- For small $t > 0$, the quadratic term dominates, so $f(\mathbf{x}^* + t\mathbf{v}) > f(\mathbf{x}^*)$
- Contradiction — therefore $H_f(\mathbf{x}^*)$ is NSD

A.5 Proof: Sufficient SOC (multivariate)

for interest only

Theorem. If $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $H_f(\mathbf{x}^*)$ is negative definite, then $\mathbf{x}^*$ is a strict local maximum.

Proof. Taylor at $\mathbf{x}^*$:

$$f(\mathbf{x}^* + \mathbf{h}) = f(\mathbf{x}^*) + \tfrac{1}{2}\mathbf{h}^\top H_f(\mathbf{x}^*)\mathbf{h} + o(\|\mathbf{h}\|^2).$$

- Negative definiteness: $\mathbf{h}^\top H_f(\mathbf{x}^*)\mathbf{h} < 0$ for all $\mathbf{h} \neq \mathbf{0}$
- For small $\mathbf{h} \neq \mathbf{0}$, the quadratic form dominates the remainder
- So $f(\mathbf{x}^* + \mathbf{h}) < f(\mathbf{x}^*)$

Connection to concavity: f concave $\Leftrightarrow$ $H_f(\mathbf{x})$ NSD for all $\mathbf{x}$ (Section 4.2; proof A.4).

A.6 Concave positive implies log-concave

for interest only

Theorem 1. Concave and positive $\Rightarrow$ log-concave.

Proof. If $f > 0$ and f concave, then $\ln f$ is concave.

For $\lambda \in [0, 1]$:

$$\ln f(\lambda x + (1-\lambda)y) \geq \ln[\lambda f(x) + (1-\lambda)f(y)]$$

by concavity of f and monotonicity of ln. Since ln is concave:

$$\ln[\lambda f(x) + (1-\lambda)f(y)] \geq \lambda \ln f(x) + (1-\lambda) \ln f(y).$$

Thus $\ln f(\lambda x + (1-\lambda)y) \geq \lambda \ln f(x) + (1-\lambda) \ln f(y)$.

A.6 Log-concave implies quasiconcave

for interest only

Theorem 2. Log-concave $\Rightarrow$ quasiconcave.

Proof. If $\ln f$ concave, then $f = e^{\ln f}$ is quasiconcave (composition of increasing concave with concave).

Specifically: if $\ln f(\lambda x + (1-\lambda)y) \geq \lambda \ln f(x) + (1-\lambda)\ln f(y)$, then

$$f(\lambda x + (1-\lambda)y) \geq f(x)^{\lambda} f(y)^{1-\lambda} \geq \min\{f(x), f(y)\}.$$

A.7 Additional exercise: Multivariate critical points

for interest only

Exercise. Find critical points of $f(x, y) = x^3 + y^3 - 3xy$.

Solution.

- $\nabla f = (3x^2 - 3y, 3y^2 - 3x) = (0, 0)$
- From first: $y = x^2$
- Substitute: $x^4 - x = 0$, so $x(x^3 - 1) = 0$
- Solutions: $(0, 0)$ and $(1, 1)$

Hessian at $(1, 1)$: $H = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$, eigenvalues $3, 9 > 0$ (local min).

At $(0, 0)$: $H = \begin{bmatrix} 0 & -3 \\ -3 & 0 \end{bmatrix}$, determinant $-9 < 0$ (saddle).

A.7 Additional exercise: Borderline SOC

for interest only

Exercise. Analyze $f(x) = -x^4$ at $x = 0$ where SOC with $f'' < 0$ fails.

Solution.

- $f'(0) = 0$, $f''(0) = 0$, $f'''(0) = 0$, $f^{(4)}(0) = -24 < 0$

Fourth-order Taylor: $f(h) = -h^4 < 0 = f(0)$ for $h \neq 0$.

Strict local maximum.

This shows higher-order terms can determine optimality when SOC is inconclusive.